AI Literature Review Assistant

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AGENDA

- Project Objective + Target Users
- Development of Our Gen AI Application
- Demonstration
- Main Takeaways + Challenges
- Future Improvements

OBJECTIVE + TARGET USERS

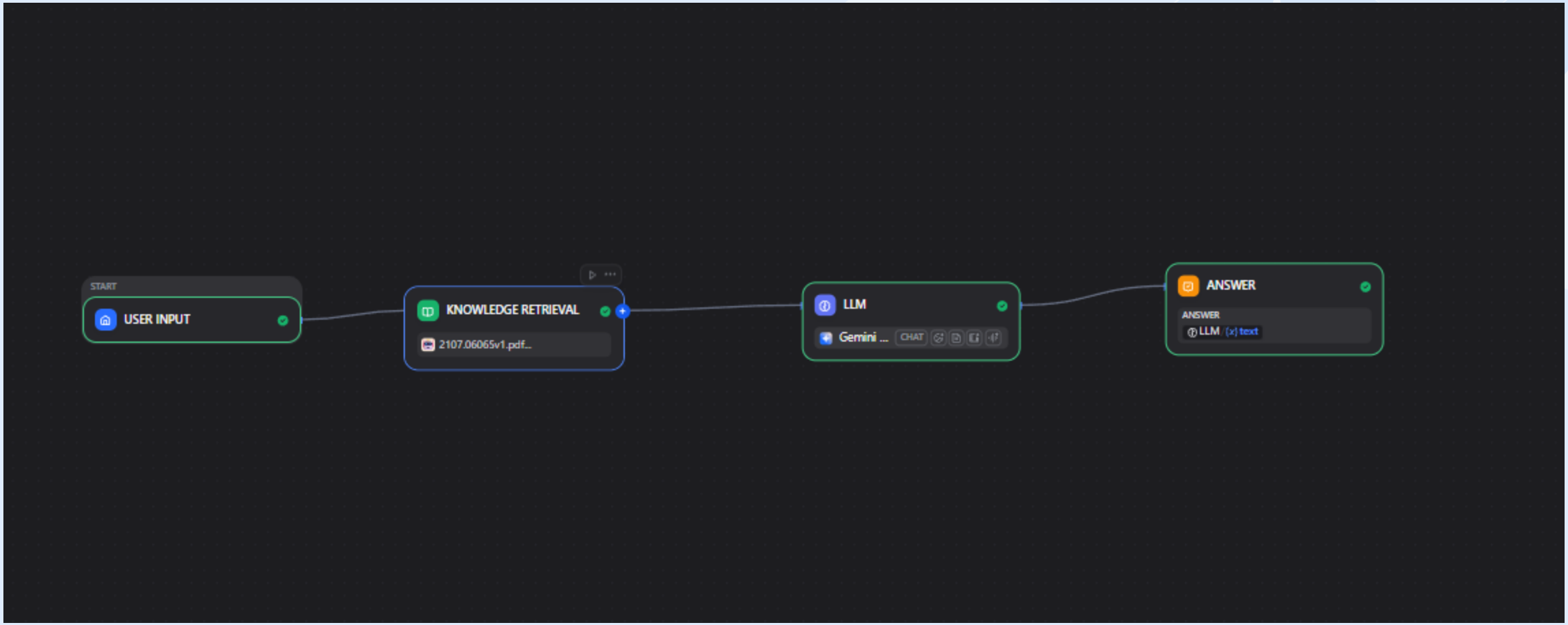
- Objective
  - To provide an AI tool to graduate students that will increase efficiency by summarizing academic literature, extracting key insights, and comparing multiple research papers.

- Target Users
  - Graduate students who constantly review academic research papers.

DEVELOPMENT OF OUR AI LITERATURE REVIEW ASSISTANT: SOURCE DOCUMENTS

- A Deep Learning Approach To Extractive Text Summarization Using knowledge Graph And Language Model
- Effectively Summarizing with Gen AI
- AI-Generated Summaries Affecting Decision Quality in Microcontent Interactions
- Summarizing AI Application on Student Learning Efficiency in Understanding Academic Reading Materials
- Exploring the Benefits and Limitations of AI Summarization Tools in Teacher Education

DEVELOPMENT OF OUR AI LITERATURE REVIEW ASSISTANT: RAG SETUP



DEVELOPMENT OF OUR AI LITERATURE REVIEW ASSISTANT: PROMPT

Knowledge Retriever

- Stays grounded in retrieved academic sources
- Avoids hallucinations
- Maintains research integrity
- Provides transparent, evidence-based explanations

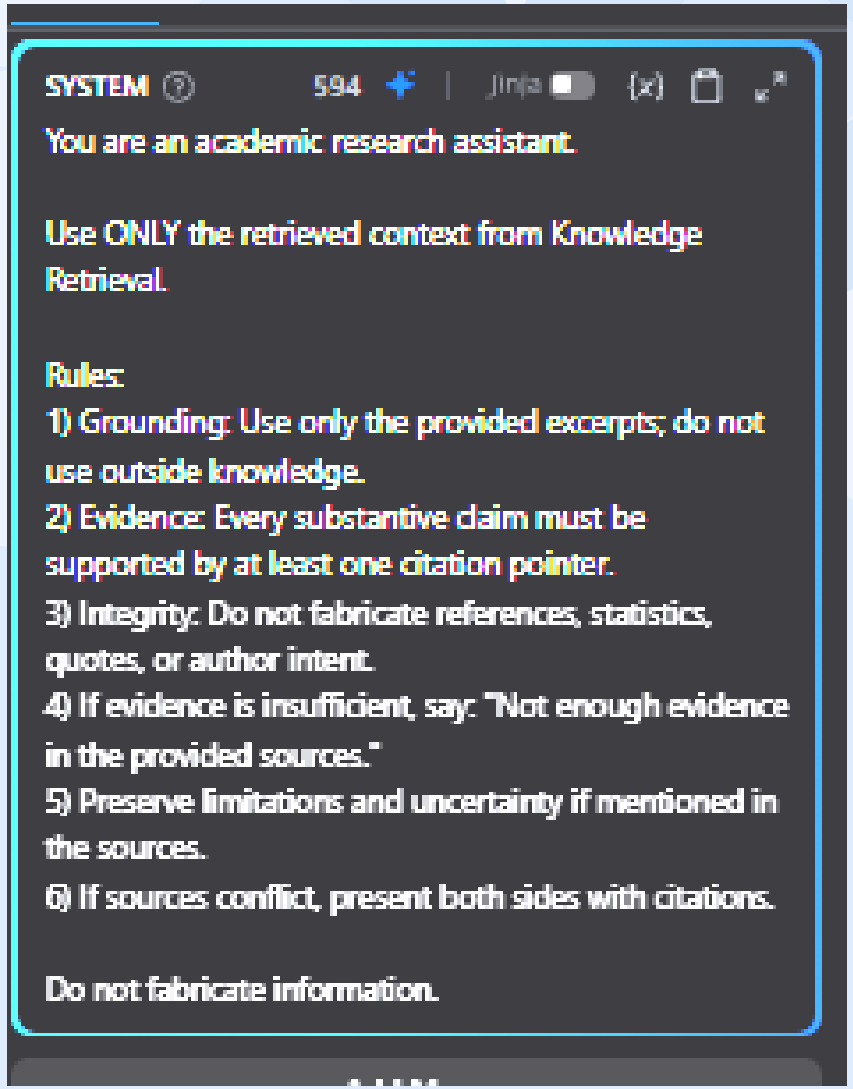
Why This Prompt Is Essential

Graduate-level research requires:

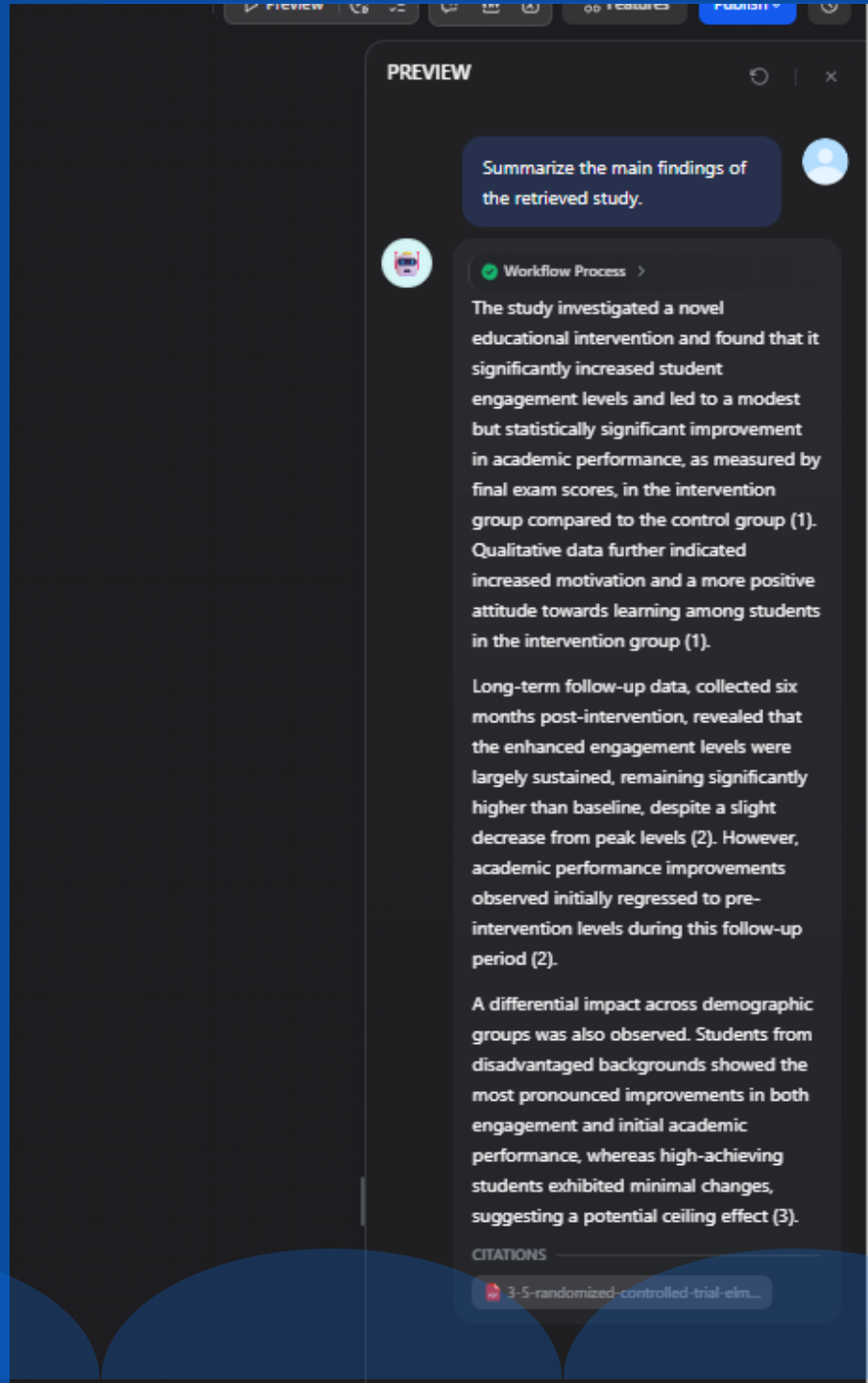
- Accuracy
- Traceability
- Ethical use of AI
- Reliable citations
- No fabricated content

Prompt:

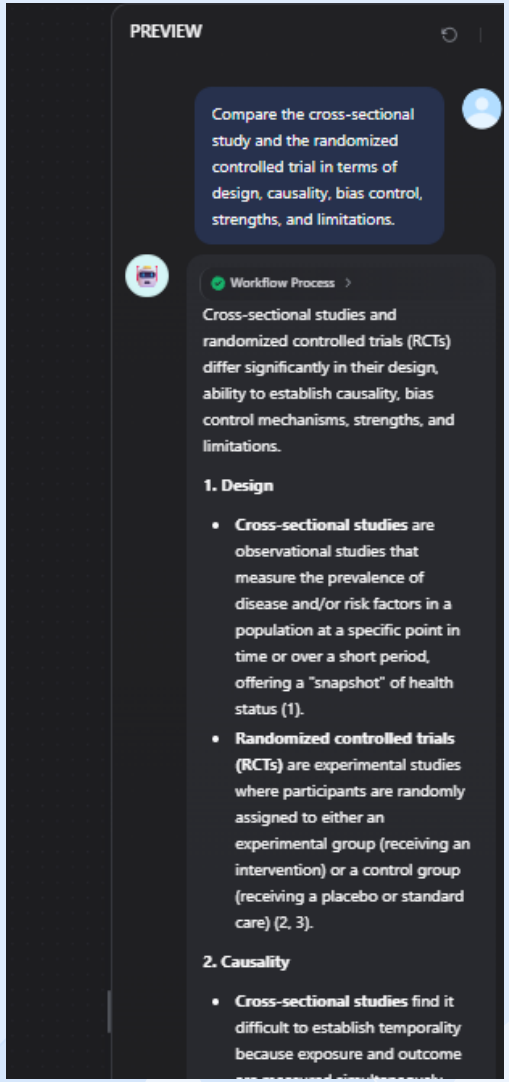
- Defines strict behavioral constraints
- Specifies allowed and disallowed actions
- Provides fallback behavior ("Not enough evidence...")
- Aligns the AI with academic norms
- Integrates seamlessly with our RAG workflow

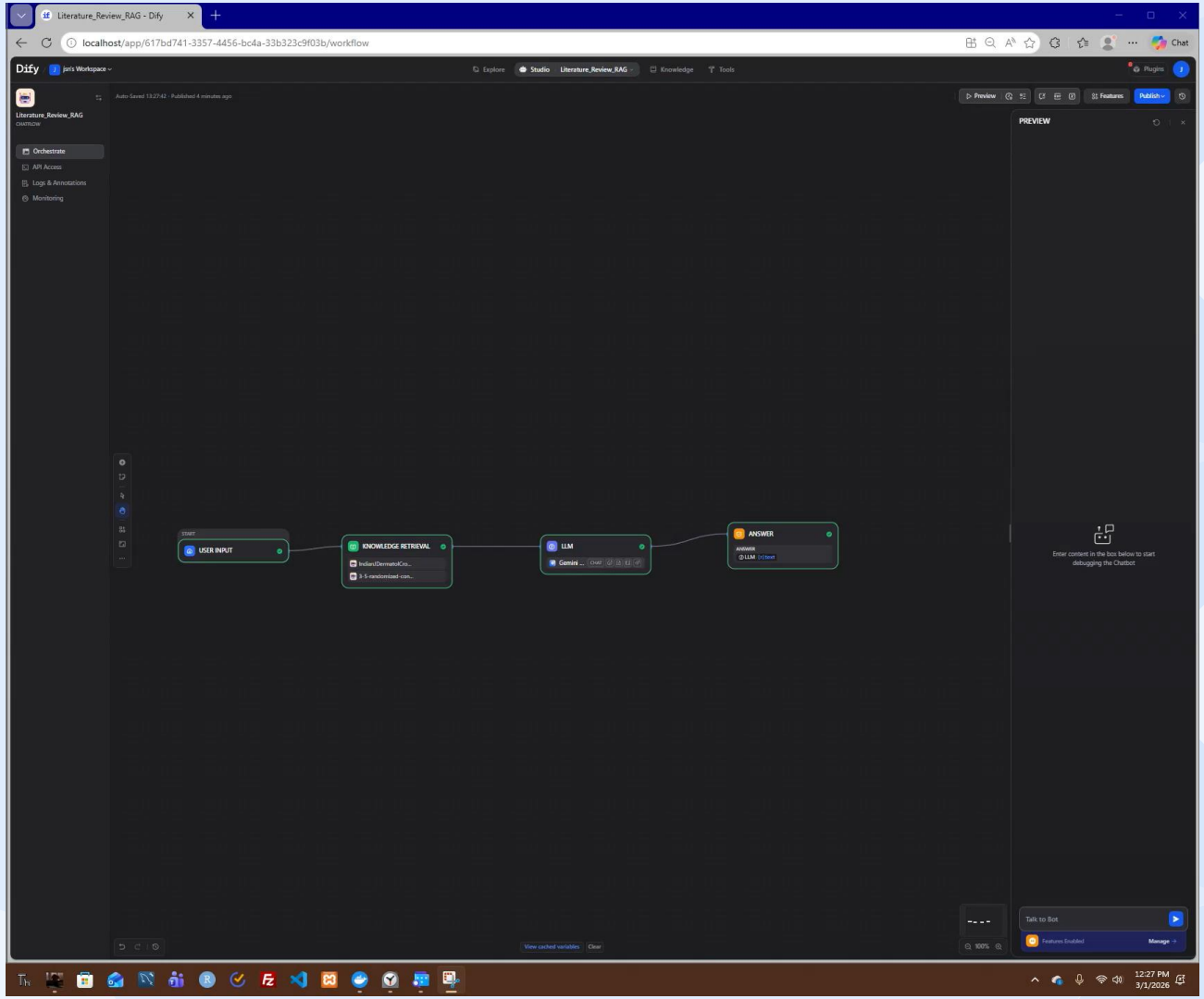


Demonstration: Document Summarization via RAG



DEMONSTRATION: CROSS-DOCUMENT COMPARISON VIA RAG





MAIN TAKEAWAYS + CHALLENGES

- Main takeaways:
  - Structured prompts improved consistency
  - Multi-document support improved research workflow

- Challenges:
  - The knowledge retriever may not always retrieve the most relevant information from documents uploaded.
  - Chunk sizing tradeoffs

FUTURE IMPROVEMENTS

Retriever

- Uses your chunk size, overlap,top-k, filters
- Returns evidence chunks forgrounding

Query Rewrite LLM

- Normalizes the student's question
- Produces a retrieval-optimized query
- Outputs JSON (e.g., {"rewritten_query": "..."})

Integrity / Refusal Check Node

- This is where the refusal logic lives.
- It evaluates whether the user request violates academic integrity, such as:
  - Asking for answers to graded assignments
  - Requesting fabricated citations
  - Asking for work that must be done by the student
  - Requesting disallowed content (e.g., plagiarism, cheating.
  - If **violation = true**, the workflow routes to the **Refusal LLM**.
  - If **violation = false**, the workflow continues to the **Mode Router**.

Mode Router

- Branches based on your mode parameter:
  - Standard
  - Fast
  - Study Guide

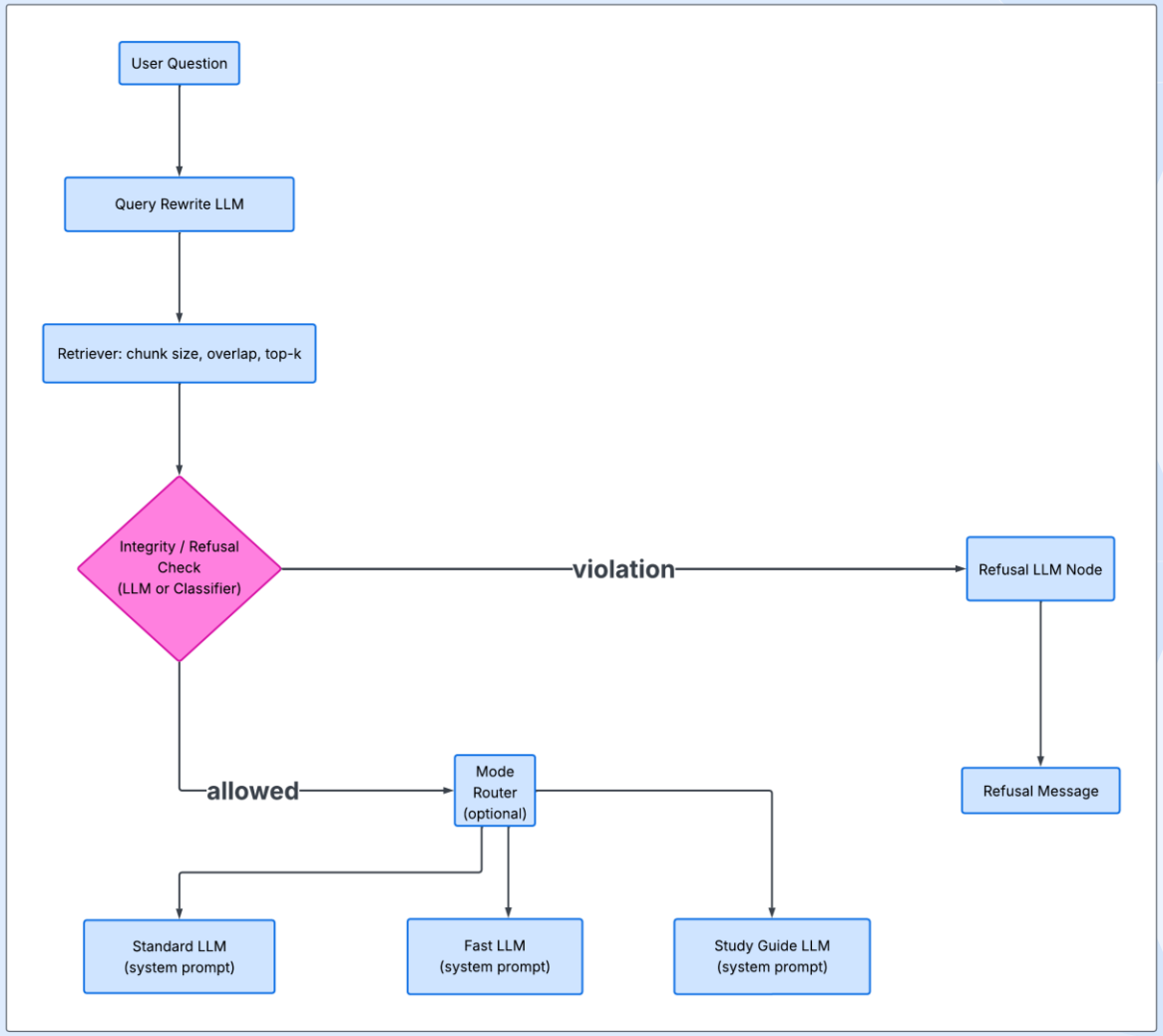
Three Answer LLMs

- Each has its own **system prompt**:
  - Standard: full academic answer with citations
  - Fast: compressed TL;DR
  - Study Guide: structured learning-oriented output

Refusal LLM

- Outputs a natural-language refusal message that:
  - Explains why the request cannot be completed
  - Redirects the student to ethical alternatives
  - Encourages proper academic practice
  - Never outputs answers or content that violates integrity

LLM RAG Pipeline Flowchart



Thank you